Last revised date : 2026-04-28
MSDS No. : AA00566-2100002085

Safety Data Sheet(SDS)

1. Identification of the substance/mixture and of the company/undertaking

1) Product identifier : ETHYL ACRYLATE

2) Relevant identified uses of the substance or mixture and uses advised against

○ Relevant identified uses

Others (Textiles sizing agents, paints and inks, glue, flexible resin, non-woven binder)

○ Uses advised against

Prohibited from use other than recommended

3) Supplier information

○ Company name [Manufacture]

Company : LG Chem, Ltd.
Address : 58, Yeosusandan 4-ro, Yeosu-si, Jeollanam-do, Republic of Korea

Emergency number : 080-880-0454

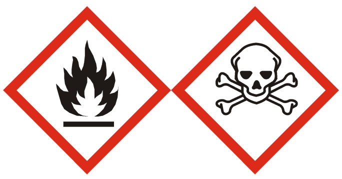
2. HAZARD IDENTIFICATION

1) Hazard classification

- Flammable liquids Category 2
- Acute toxicity(Oral) Category 4
- Acute toxicity(Dermal) Category 4
- Acute toxicity(Inhalation:Vapours) Category 3
- Skin corrosion/irritation Category 2
- Serious eye damage/eye irritation Category 2
- Skin sensitization Category 1
- Specific target organ toxicity single exposure Category 3(Respiratory tract irritation)
- Hazardous to the aquatic environment, long-term (chronic) Chronic 3

2) Allocation label elements

Hazard pictograms



Signal word

- DANGER

Hazard statements

H225 Highly flammable liquid and vapour
H302 Harmful if swallowed
H312 Harmful in contact with skin
H315 Causes skin irritation
H317 May cause an allergic skin reaction
H319 Causes serious eye irritation
H331 Toxic if inhaled
H335 May cause respiratory irritation
H412 Harmful to aquatic life with long lasting effects

Precautionary statements

- Prevention

P210 Keep away from heat, hot surfaces, sparks, open flames and other ignition sources. No smoking.
P233 Keep container tightly closed.
P240 Ground and bond container and receiving equipment.
P241 Use facilities such as explosion-proof electricity, ventilation and lighting.
P242 Use nonsparking tools.
P243 Take action to prevent static discharges.
P261 Avoid breathing mist/vapours/spray.
P264 Wash mouse,eye,skin thoroughly after handling.
P270 Do not eat, drink or smoke when using this product.
P271 Use only outdoors or in a wellventilated area.
P272 Contaminated work clothing should not be allowed out of the workplace.
P273 Avoid release to the environment.
P280 Wear protective gloves/protective clothing/eye protection/face protection.

- Response

P301+P312 If you swallow: If you feel uncomfortable, get medical institutions and doctor's consultation.
P302+P352 If you get on your skin: Wash with a large amount of water.
P303+P361+P353 IF ON SKIN (or hair): Take off immediately all contaminated clothing. Rinse skin with water [or shower].
P304+P340 IF INHALED: Remove person to fresh air and keep comfortable for breathing.
P305+P351+P338 IF IN EYES: Rinse cautiously with water for several minutes. Remove contact lenses, if present and easy to do. Continue rinsing.
P311 Get medical institutions and doctors' consultation.
P312 If you feel uncomfortable, receive medical institutions and doctors' consultation.
P321 If you eat or inhale the substance, do not take artificial breath in the oral cavity, and use appropriate breathing medical equipment.,When contacting with substances, make a treatment such as rinsing the skin and eyes in the water flowing more than 20 minutes.
P330 Rinse mouth.
P332+P313 If skin irritation occurs: Get medical advice/attention.
P333+P313 If skin irritation or a rash occurs: Get medical advice/attention.

P337+P313 If eye irritation persists: Get medical advice/attention.

P362+P364 Take off contaminated clothing and wash it before reuse.

P370+P378 In case of fire: Use carbon dioxide,Water,Alcoholic to extinguish.

- Storage

P403+P233 Store in a wellventilated place. Keep container tightly closed.

P403+P235 Store in a wellventilated place. Keep cool.

P405 Store locked up.

- Disposal

P501 Dispose of contents and containers according to the legislation of the waste

3) Other hazards

- No data available

○ Product NFPA Level

Health	Flammability	Reactivity
2	2	1

(※ 0 = Stable , 1 = Low , 2 = Medium , 3 = High , 4 = Very High)

3. Composition/Information on ingredients

Components	Common name	CAS No.	PCT(wt%)
ETHYL ACRYLATE	ETHYL ACRYLATE	140-88-5	100

4. FIRST AID MEASURES

1) Following eye contact

- Seek immediate medial assistance.
- In case of contact with substance, immediately flush skin or eyes with running water for at least 20 minutes.

2) Following skin contact

- For minor skin contact, avoid spreading material on unaffected skin.
- Seek immediate medial assistance.
- For hot product, immediately immerse in or flush the affected area with large amounts of cold water to dissipate heat.
- In case of burns, immediately cool the affected area with cold water for as long as possible and do not remove clothing that is stuck to the skin.
- In case of contact with substance, immediately flush skin or eyes with running water for at least 20 minutes.
- Wash skin with soap and water.
- Remove and isolate contaminated clothing and shoes.

3) Following inhalation

- Do not perform artificial respiration by mouth-to-mouth method, but use appropriate respiratory medical equipment.
- Seek immediate medial assistance.
- Keep victim warm and quiet.
- Move to fresh air.
- Ensure that medical personnel are aware of the material(s) involved and take precautions to protect themselves.
- Effects of contact or inhalation may be delayed.
- Administer oxygen if breathing is difficult.
- Give artificial respiration if victim is not breathing.

4) Following ingestion

- Do not perform artificial respiration by mouth-to-mouth method, but use appropriate respiratory medical equipment.
- Seek immediate medial assistance.

5) Advice to physician

- Ensure that medical personnel are aware of the material(s) involved and take precautions to protect themselves.
- Effects of contact or inhalation may be delayed.

5. FIRE FIGHTING MEASURES

1) Suitable (and unsuitable) extinguishing media

○ Suitable extinguishing media

- CO2.
- Dry chemical.
- For mixtures containing alcohol or polar solvent: Alcohol-resistant foam.
- Water spray.
- Use alcohol foam, carbon dioxide, or water spray when fighting fires involving this material.
- Regular foam.
- Use dry sand or earth to smother fire.

○ Unsuitable extinguishing media

- High-pressure water.
- Direct water.

2) Special hazards arising from the substance or mixture

○ Pyrolytic product

- During burning, irritating and highly toxic gases may be produced by thermal decomposition or combustion.

○ Risk of fire and explosion

- Containers may explode when heated.

- When heated, vapors may form explosive mixtures with air: indoors, outdoors and sewers explosion hazards.
- May violently polymerize and result in fire and explosion.
- HIGHLY FLAMMABLE: Will be easily ignited by heat, sparks or flames.
- Runoff may create fire or explosion hazard.
- Vapor explosion hazard indoors, outdoors or in sewers.
- Can form explosive mixtures at temperatures at or above the flashpoint.
- Vapors may form explosive mixtures with air.
- Vapors may travel to source of ignition and flash back.

○ Other

- Fire may produce irritating, corrosive and/or toxic gases.
- May cause toxic effects if inhaled or absorbed through skin.

3) Special protective equipment for firefighters

- Rescuers should put on appropriate protective gear.
- Cautions ; Most of liquids are lighter than water.
- Most vapors are heavier than air. They will spread along ground and collect in low or confined areas (sewers, basements, tanks).
- Dike fire-control water for later disposal; do not scatter the material.
- Move containers from fire area if you can do it without risk.
- Evacuate area and fight fire from a safe distance.
- Fire involving Tanks: For massive fire, use unmanned hose holders or monitor nozzles; if this is impossible, withdraw from area and let fire burn.
- Fire involving Tanks: Cool containers with flooding quantities of water until well after fire is out.
- Fire involving Tanks: Withdraw immediately in case of rising sound from venting safety devices or discoloration of tank.
- Fire involving Tanks: Fight fire from maximum distance or use unmanned hose holders or monitor nozzles.
- Fire involving Tanks: ALWAYS stay away from tanks engulfed in fire.

6. ACCIDENTAL RELEASE MEASURES

1) Health considerations and protective equipment

- Do not touch or go near exposed material.
- The very fine particles can cause a fire or explosion, eliminate all ignition sources.
- ELIMINATE all ignition sources (no smoking, flares, sparks or flames in immediate area).
- All equipment used when handling the product must be grounded.
- Clean up spills immediately, observing precautions in Protective Equipment section.
- Stop leak if you can do it without risk.
- Do not touch damaged containers or spilled material unless wearing appropriate protective clothing.
- A vapor suppressing foam may be used to reduce vapors.

- Cover with plastic sheet to prevent spreading.
- Please note that materials and conditions to be avoided.

2) Environmental precautions

- Runoff may cause pollution.
- Prevent entry into waterways, sewers, basements or confined areas.
- Keep out of waterways.

3) For cleaning up

- Absorb or cover with dry earth, sand or other non-combustible material and transfer to containers.
- Large Spill: Dike far ahead of liquid spill for later disposal.
- Absorb spill with inert material (e.g., dry sand or earth), then place in a chemical waste container.
- Small Spill: Absorb with earth, sand or other non-combustible material and transfer to containers for later disposal.

- Dike and collect water used to fight fire.
- Absorb the liquid and scrub the area with detergent and water.
- Use clean non-sparking tools to collect absorbed material.

7. HANDLING AND STORAGE

1) Precautions for safe handling

- Avoid breathing vapors from heated material.
- Loosen closure cautiously before opening.
- CAUTION: High temperature.
- Handling refer to engineering control/personal protection section.
- All equipment used when handling the product must be grounded.
- DO NOT PRESSURIZE, CUT, WELD, BRAZE, SOLDER, DRILL, GRIND, OR EXPOSE SUCH CONTAINERS TO HEAT, FLAME, SPARKS, STATIC ELECTRICITY, OR OTHER SOURCES OF IGNITION.
- Caution: Heat.
- Follow all SDS/label precautions even after container is emptied because they may retain product residues.
- Avoid prolonged or repeated contact with skin.
- Measure atmospheric oxygen concentration and ventilate the area during the operation since low-closed area can cause oxygen deficiency.
- Do not enter storage area unless adequately ventilated.
- Use care in handling/storage.
- Please note that materials and conditions to be avoided.
- Use only in a well-ventilated area.

2) Conditions for safe storage (including any incompatibilities)

- Empty drums should be completely drained, properly bunged, and promptly returned to a drum reconditioner, or properly disposed of.
- Keep away from food and drinking water.

8. EXPOSURE CONTROLS AND PERSONAL PROTECTION

1) Chemical exposure limits, Biological exposure standard

Components	Occupational exposure limits	ACGIH	Biological standard
ETHYL ACRYLATE	TWA : 5.0ppm STEL : Not applicable	TWA : ppmmg/m3 STEL : ppmmg/m3	Not applicable

2) Appropriate engineering controls

- Use process enclosures, local exhaust ventilation, or other engineering controls to control airborne levels below recommended exposure limits.
- If user operations generate dust, fume, or mist, use ventilation to keep exposure to airborne contaminants below the exposure limit.
- Facilities storing or utilizing this material should be equipped with an eyewash facility and a safety shower.

3) Personal protection equipment

○ Respiratory protection

- Wear breathing protection, which needs a confirmation from the Korea Occupational Safety and Health Agency.

- In case of insufficient oxygen (<19.6%), wear a supplied air mask or self-contained respirator.

○ Eye protection

- Wear suitable protective goggles and face shields.

○ Hand protection

- Wear suitable protective gloves.

○ Body protection

- Wear suitable protective clothing.

9. PHYSICAL AND CHEMICAL PROPERTIES

Appearance	Clear Liquid
Physical state	Liquid
Colour	Colorless
Odour	pungent smell
Odour threshold	pungent smell
pH	No data available
Melting point/freezing point	-71 ℃
Initial boiling point and boiling range	100℃
Flash point	8℃
Evaporation rate	3.8 ㎪(20℃)
Flammability(solid, gas)	No data available
Upper/lower flammability or explosive limits	14 % / 1.4 %
Vapour pressure	3.8 ㎪(20℃)

Solubility(ies)	No data available
Vapour density	No data available
Relative density	No data available
n-octanol/water partition coefficient	1.32
Auto ignition temperature	345 ℃
Decomposition temperature	No data available
Viscosity	0.63㎟/s
Molecular weight(mass)	100.12 g/mol

10. STABILITY AND REACTIVITY

1) Stability and hazardous reactivity

- Containers may explode when heated.
- When heated, vapors may form explosive mixtures with air: indoors, outdoors and sewers explosion hazards.
- May violently polymerize and result in fire and explosion.
- HIGHLY FLAMMABLE: Will be easily ignited by heat, sparks or flames.
- Runoff may create fire or explosion hazard.
- Vapor explosion hazard indoors, outdoors or in sewers.
- Can form explosive mixtures at temperatures at or above the flashpoint.
- Vapors may form explosive mixtures with air.
- Vapors may travel to source of ignition and flash back.
- Fire may produce irritating and/or toxic gases.
- Fire may produce irritating, corrosive and/or toxic gases.
- May cause toxic effects if inhaled or absorbed through skin.

2) Conditions to avoid

- Containers may explode when heated.
- HIGHLY FLAMMABLE: Will be easily ignited by heat, sparks or flames.
- Runoff may create fire or explosion hazard.
- Vapor explosion hazard indoors, outdoors or in sewers.
- Ignition source(heat, spark, flame, etc.).
- Heat, contamination.
- Heat.
- Vapors may form explosive mixtures with air.
- Vapors may travel to source of ignition and flash back.
- Fire may produce irritating, corrosive and/or toxic gases.
- May cause toxic effects if inhaled or absorbed through skin.

3) Incompatible materials

- Combustibles, reducing material.
- Containers may explode when heated.

- HIGHLY FLAMMABLE: Will be easily ignited by heat, sparks or flames.
- Runoff may create fire or explosion hazard.
- Vapor explosion hazard indoors, outdoors or in sewers.
- Vapors may form explosive mixtures with air.
- Vapors may travel to source of ignition and flash back.
- Fire may produce irritating, corrosive and/or toxic gases.
- May cause toxic effects if inhaled or absorbed through skin.

4) Hazardous decomposition products
- Corrosive/toxic fume.
- Ignition source(heat, spark, flame, etc.).
- Heat.
- Irritating and/or toxic gas.
- Irritating, corrosive and/or toxic gas.
- During burning, irritating and highly toxic gases may be produced by thermal decomposition or combustion.

11. TOXICOLOGICAL INFORMATION

1) Exposure route information

○ Inhalation
- May cause respiratory irritation
- Toxic if inhaled

○ Skin Contact
- May cause an allergic skin reaction
- Causes skin irritation
- Harmful in contact with skin

○ Eye Contact
- Causes serious eye irritation

○ Ingestion
- Harmful if swallowed

2) Health hazard information

○ Acute toxicity

● Acute toxicity(Oral)

Korea Ministry of Environment(Category 4 : 500mg/kg) - Acute toxic substances for humans(2023-1-1152)

● Acute toxicity(Dermal)

Korea Ministry of Environment(Category 4 : 1100mg/kg) - Acute toxic substances for humans(2023-1-1152)

● Acute toxicity(Inhalation:Gases)

Korea Ministry of Environment(Category 3 : 700ppm) - Acute toxic substances for humans(2023-1-

1152)

- Acute toxicity(Inhalation:Vapours)

  LC50 5.8 ㎎/ℓ 4 hr Test species: Rat (2180 ppm 4h, OECD TG 403), Source: ECHA

- Acute toxicity(Inhalation:Dust/mist)

  No data available

○ Skin corrosion/irritation

Korea Ministry of Environment(Category 2) - Acute toxic substances for humans(2023-1-1152)

○ Serious eye damage/eye irritation

Korea Ministry of Environment(Category 2) - Acute toxic substances for humans(2023-1-1152)

○ Respiratory sensitization

Causes skin sensitization, CBA mouse, OECD TG 429

○ Skin sensitization

Korea Ministry of Environment(Category 1) - Acute toxic substances for humans(2023-1-1152)

○ Carcinogenicity

2B (IARC)
A4 (ACGHI)
2 (Ministry of Employment and Labor notice)

○ Germ cell mutagenicity

In vitro microbial reverse mutation assays and in vivo mammalian micronucleus assays showed negative results., Source: OECD SIDS

○ Reproductive toxicity

In a reproductive toxicity test using rats, no significant effects were observed other than a decrease in fetal weight. NOAEL = 100 ppm (0.41 mg/L) (maternal, fetal toxicity), Source: OECD SIDS

○ Specific target organ toxicity single exposure

Korea Ministry of Environment(Category 3(Respiratory tract irritation)) - Acute toxic substances for humans(2023-1-1152)

○ Specific target organ toxicity repeated exposure

Repeated oral toxicity tests using rats showed no deaths or significant adverse effects. NOAEL = 55 mg/kg bw/day OECD TG 408, Source: ECHA

○ Aspiration hazard

No data available

12. ECOLOGICAL INFORMATION

1) Aquatic toxicity

- Fish

  LC50 4.6 mg/ℓ 96 hr Other (rainbow trout), Source: OECD SIDS

- Crustacea

EC50 7.9 ㎎/ℓ 48 hr Daphnia magna, Source: OECD SIDS

- Aquatic algae

  EC50 2.02 ㎎/ℓ 96 hr Selenastrum capricornutum, Source: ECHA

2) Persistence and degradation

- n-octanol water partition coefficient

  1.32 log Kow, Source: ICSC

- Degradation

  No data available

- Biodegradation

  57.3 % 28 day (OECD SIDS 301 D), Source: OECD SIDS

3) Bioaccumulative potential

No data available

4) Mobility in soil

3.9 ~ 85 Koc (EPA OTS 796.2750 Sediment and Soil Adsorption Isotherm), Source: ECHA

5) Other adverse effects

NOECDaphnia magna, 21d = 0.19 mg/L

13. DISPOSAL CONSIDERATIONS

1) Disposal methods

- Recyclables must be collected in portable bins and disposed of legally according to national regulations.

2) Precautions (including disposal of contaminated container of package)

- Avoid direct spillage to rivers, lakes, soil and drains.

14. TRANSPORT INFORMATION

1) UN No. : 1917

2) Proper shipping name : ETHYL ACRYLATE, STABILIZED

3) Class or division : 3

4) Packing group : II

5) Marine pollutant : Not applicable

6) Special safety response for transportation or transportation measure :

Emergency measures in case of fire : F-E

Emergency measures in the effluent : S-D

- ADR

  · Tunnel restriction code : D/E

- IMDG

· Marine pollutant : Not applicable

- Air transport(IATA)

  · UN No. : 1917

  · Proper shipping name : ETHYL ACRYLATE, STABILIZED

  · Class or division : 3

  · Packing group : II

15. REGULATORY INFORMATION

1) Occupational Safety and Health Act in Korea

   Environment measure substance(more than 1%), Hazardous Substances(Substances Subject to Process Safety Reporting), Hazardous SubstancesRequiring Management(more than 1%), Special medical examination material(more than 1%), Substance exposure limits

2) Toxic Chemical Control Act in Korea

   Acute toxic substances for humans(more than 25%), Pollutant release and transfer register substances(more than 0.1%)

3) Safety Control of Dangerous Substances Act in Korea

   Class 4〉 First petroleum (water-insoluble)(Designated quantity:200ℓ)

4) Wastes Control Act in Korea

   Designated waste (Waste toxic substances)

   -In case of disposal, it must be disposed of in accordance with Article 13 of the Waste Management Act.

5) Other regulations in KOREA and Abroad regulations

   ● ETC regulation

   - Not applicable
   - Not applicable

   ● Act on the registration and evaluation of chemicals

   Existing Commercial Chemical Substances

16. OTHER INFORMATION

1) Reference

2) Date of first issue : 2021-12-15

3) Revision date

   ○ Revised date count : 6

○ Last revised date : 2026-04-28

○ Last revised history : Designation as a toxic substance when EA content is 25% or higher

4) Other